Matrix Multiplication

Matrix multiplication is a fundamental operation in linear algebra. To multiply two matrices, follow these steps carefully.

Step 1: Check Dimensions
Matrix multiplication is only defined if the number of columns in the first matrix equals the number of rows in the second matrix. If A is $m \times n$ and B is $p \times q$, then $A \cdot B$ is defined only if $n = p$, and the resulting matrix will have dimensions $m \times q$.

Step 2: Multiply Row by Column
To compute the entry in row i, column j of the product $C = A \cdot B$:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

That is:

1. Take row i of the first matrix.
2. Take column j of the second matrix.
3. Multiply corresponding entries and sum them.

Step 3: Fill the Resulting Matrix
Repeat Step 2 for every row of the first matrix and every column of the second matrix.

Matrix multiplication is generally *not commutative*, meaning:

$$A \cdot B \neq B \cdot A$$

Example 1: Consider two matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

We want to compute the product $C = A \cdot B$. Each entry C_{ij} is computed as the **dot product of the i-th row of A and the j-th column of B**.

$$\begin{aligned} C_{11} &= (1 \cdot 5) + (2 \cdot 7) = 5 + 14 = 19 \\ C_{12} &= (1 \cdot 6) + (2 \cdot 8) = 6 + 16 = 22 \\ C_{21} &= (3 \cdot 5) + (4 \cdot 7) = 15 + 28 = 43 \\ C_{22} &= (3 \cdot 6) + (4 \cdot 8) = 18 + 32 = 50 \end{aligned}$$

Therefore, the resulting matrix is:

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Example 2: Consider:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

We want to compute $C = A \cdot B$. Each entry C_{ij} is the **dot product of the i-th row of A and the j-th column of B**.

$$C_{11} = (1 \cdot 3) + (0 \cdot 2) + (2 \cdot 1) = 3 + 0 + 2 = 5$$

$$C_{12} = (1 \cdot 1) + (0 \cdot 1) + (2 \cdot 0) = 1 + 0 + 0 = 1$$

$$C_{13} = (1 \cdot 2) + (0 \cdot 0) + (2 \cdot 1) = 2 + 0 + 2 = 4$$

$$C_{21} = (-1 \cdot 3) + (3 \cdot 2) + (1 \cdot 1) = -3 + 6 + 1 = 4$$

$$C_{22} = (-1 \cdot 1) + (3 \cdot 1) + (1 \cdot 0) = -1 + 3 + 0 = 2$$

$$C_{23} = (-1 \cdot 2) + (3 \cdot 0) + (1 \cdot 1) = -2 + 0 + 1 = -1$$

Therefore, the resulting matrix is:

$$C = A \cdot B = \begin{bmatrix} 5 & 1 & 4 \\ 4 & 2 & -1 \end{bmatrix}$$

Note: Now $B \cdot A$ is **not defined** because B is 3×3 and A is 2×3. This shows that **matrix multiplication is not commutative**.